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Minimizing Temperature Nonuniformity by Optimal Arrangement of Hotspots in Vertically Stacked Three-Dimensional Integrated Circuits

The semiconductor packaging technologies have seen its growth from multichip module (MCM), system in package (SiP), system on chip (SoC) to the heterogeneous integration of the MCM. Thermal management of multichip vertically integrated systems poses additional constraints and limitations beyond those for single chip modules. Three-dimensional-integrated circuits (3D ICs) technology is a potential approach for next-generation semiconductor packaging technologies. A 3D IC is formed by vertical interconnection of multiple substrates containing active devices which offer reduced die footprint and interconnect length. This paper discusses the optimal arrangement of two hotspots on each layer of a two-die stacked 3D IC. An analytical heat transfer model for prediction of three-dimensional temperature field of a 3D IC based on the solution of governing energy equations has been developed and used for this study. The model is subject to adiabatic boundary conditions at the walls except for the bottom wall which is subject to convective boundary condition. A feed-forward back propagation artificial neural network (ANN) is employed for obtaining the functional relationship between the location of the hotspots and the objectives. Genetic algorithm is employed for solving two nonconflicting objective functions subject to set of constraints. The first objective aims to minimize the maximum temperature on both layers, and the second objective aims to achieve temperature uniformity in the layers. The results of the optimization study are expected to provide recommendations on the design guidelines for arranging hotspots on vertically stacked substrates. [DOI: 10.1115/1.4047471]

Introduction

Three-dimensional integrated circuits (3D ICs) possess the potential to upgrade chip performance, functionality, and packaging density if they can be thermally managed. A 3D-IC is typically envisioned as a stack of tiers of active chips connected down to package substrates using through-silicon-via (TSVs). By replacing the two-dimensional system-on-chips (2D SoCs) with three-dimensional system-on-chips (3D SoCs), a drastic decrease in interconnect length is achieved which directly results in higher system speed [1]. Pangracious et al. [1] shows that by replacing a 2D SoC with surface area of A and thickness of l_v by a 3D SoC with n tiers and same total surface area, one can reduce the maximum Manhattan wire length (Manhattan wiring is often used to represent a programmable logic array) from $2\sqrt{A}$ to $2\sqrt{\frac{A}{n}} + (n+1)l_v$. Higher power efficiency, increased bandwidth, reduced latency, lower area, and weight are the advantages of 3D SoCs compared to 2D SoCs, resulted from shorter interconnections. Also in comparison to 2D systems, 3D system is more power efficient. By increasing the power density, the temperature of the substrates closer to the package increases and results in degraded performance and higher temperature gradient that expedite wear out mechanisms.

Thermal performance of vertical integrated circuits (VICs) were explored in detail by the researchers. Kleiner et al. [2] studied the effect of polyimide and silicon layer thicknesses and

thermal conductivity on self-heating metal-oxide semiconductor field effect transistors (MOSFETs) numerically and experimentally. Sienski et al. [3] studied the impact of multichip module (MCM) substrates direct and lateral thermal conductivities on thermal performance of 3D-ICs. They compared air-cooling technique with spray-cooling technique in sophisticated electronics. A detailed thermal performance analysis of 3D-ICs under different integration schemes was presented by Souri and Saraswat [4]. They studied the effect of TSVs and power consumption on the thermal performance considering active devices and interconnect resistance. A mathematical model was developed by Geer et al. [5] to solve the steady-state conduction heat transfer in multiple rectangular domains. The developed model is general and involves the effect of interface resistance and spatially varying heat generation rate. The models have been applied to a 3D stacked system with a specified internal volumetric heat generation. Goplen and Sapatnekar [6] studied the effect of thermal vias position and density on the thermal performance of 3D-ICs. The authors calculated the thermal vias position resulting in lower module maximum temperature and minimum thermal gradients by employing a finite element analysis. Choobineh and Jain [7] calculated the three-dimensional temperature profile in a 3D-IC with equal and unequal sized dies analytically. The authors used their model to compare the thermal behavior of unequally sized die stacks with uniformly sized die stacks. In a separate study, Choobineh and Jain [8] offered a noniterative analytical approach to investigate the three-dimensional temperature field in a 3D-IC with large number of dies. They predicted the effect of interdie contact resistance on the temperature distribution and thermal performance of the 3D-IC.

In addition, Choobineh et al. [9–13] investigated interdie thermal resistance in two die stack 3D ICs both, numerically and

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Contributed by the Electronic and Photonic Packaging Division of ASME for publication in the JOURNAL OF ELECTRONIC PACKAGING. Manuscript received December 18, 2019; final manuscript received June 2, 2020; published online June 29, 2020. Assoc. Editor: Sreekant Narumanchi.

experimentally. Li et al. [14] performed a novel thermal optimization using incremental floor planning. The authors highlighted the nonoverlapping constraint and using additional binary variables to indicate overlap. Jain et al. [15] solved conflicting thermal and electrical objectives for floor planning in 3D ICs using the simulated annealing. The authors did not employ a mathematical model for the electrical and thermal objectives, rather minimized the cost function that reduced the effective wire length and temperature of the die.

Hagan and Henhaj [16] showed how to incorporate backpropagation algorithm to train artificial neural networks (ANNs). The Marquardt algorithm was also discussed and compared with the conjugate gradient algorithm and with variable learning rate. It was found that the Marquardt algorithm was very efficient when training networks to a few hundred neuron interconnection weights. Srikanth et al. [17,18] reported the results of a numerical optimization and experimental investigation of phase-change material-based composite pin fin matrix heat sink. Nondominated sorting genetic algorithm II multi-objective optimization approach was used to optimize the heat sink dimensions.

Srikanth and Balaji [19] developed heat transfer correlation for thermal performance of the heat sink as a function of heat source location and power level. Tapano et al. [20] studied the optimal arrangement of discrete heat sources in mixed convection. An empirical correlation is developed for the nondimensional maximum temperature excess (θ) in terms of a unique geometrical parameter, by taking into account the effect of surface radiation. Gosselin et al. [21] reviewed the utilization of genetic algorithms in three main families of heat transfer problems, namely, thermal system design, inverse heat transfer, and heat transfer data reduction for heat transfer correlations. Maddadi and Balaji [22] performed an optimization of the location of multiple discrete heat sources in a ventilated cavity using artificial neural networks and microgenetic algorithm. Osmolovsky et al. [23] optimally arranged the heat sources for an interposer-based 3D ICs. To optimally arrange the placement, the authors moved all dies together as one "virtual die," minimizing the terminal connections and performed the technique outlined above for each die, in multiple iterations, until no further improvement is achieved. Kumar et al. [24] performed optimal floor planning on VLSI using particle swarm optimization. The authors imposed the nonoverlapping constraint to the optimization algorithm and derived optimal location of heat sources satisfying the constraint. Ni et al. [25] demonstrated floor planning in 3D ICs and highlight the greedy placement strategy. The heaters with high heat flux are recommended to be placed closer to the convective surface.

Generally, heat sinks are designed assuming that a uniform heat flux is produced at the top surface of a microprocessors (convective surface) and the performance of the heat sink is evaluated in terms of the thermal resistance (one-dimensional) or maximum temperature rise. For efficient thermal management of microchips with multiple heat sources, uniform temperature distribution is preferred on the heat sink surface.

It is evident from the literature that one of the major challenges in designing an efficient heat sink is to minimize the temperature nonuniformity on a microprocessor surface caused by heterogeneous diverse power distribution. This work in this paper is not motivated by the necessity to develop thermal solutions which can maintain uniform temperature distribution at the heat sink's surface, rather to find optimal arrangement of the heat sources which provide a uniform temperature distribution in a double-layered 3D-IC system. Furthermore, the floor planning technique for 3D ICs available in literature lacks a mathematical model or a regression equation to characterize the location of the heat sources. Also, the nonoverlapping constraint is not imposed so efficiently in the optimization problem available in literature. This work aims to efficiently impose and characterize the nonoverlapping constraint in the optimization problem by defining a unique geometrical parameter and obtain solutions using a stochastic optimization algorithm. An analytical-numerical model coupled with

ANN and genetic algorithm was used to find the optimum position of hotspots on the substrates. The optimization was done for minimizing system temperature gradient and maximum temperature at fixed thermal boundary conditions. A convective boundary condition was used in the model to simulate the heat removed by the heat sink.

Heat Transfer Problem Description

The schematic of the geometry under study and the governing energy equations are shown in Fig. 1. The dimensions of the chip and substrate are tabulated in Table 1. The governing equations along with the boundary conditions are solved analytically as mentioned in Choobineh and Jain [8]. The convective heat transfer coefficient h value is set as $5000 \text{ W/m}^2\text{K}$ at the bottom wall of layer 1. Each hot spot generates $Q = 5 \text{ W}$ heat. The material properties of the substrate are taken as the value of pure silicon. The top surface and side walls are adiabatic. The die stack consists of two dies, each of size $a \times b$. Thickness of the i th die is c_i . The maximum value for i in this paper is two. The orientation of the x and y axis are common to each die. For brevity, x_i and y_i coordinates on each die are referred to as x and y , respectively. A heat sink is attached to the lowest die [9]. Heat transfer from the bottom-most die to the air, which typically transfers through a thermal interface material, heat spreader, heat sink, and forced air cooling, is lumped together and modeled using a combined convective boundary condition, represented as h [9]. Temperature uniformity on this surface is highly preferred for the best heat sink thermal resistance. For low-power applications where no heat sink is provided, the convective boundary condition quantifies the heat dissipated through the electrical package. Topmost die and side surfaces are assumed to be adiabatic. $Q_i(x, y)$ is the heat generation in the device plane at the top of the i th die [9]. This analysis is simplified by deriving the temperature field assuming that instead of all n heat sources, only the i *th heat source $Q_i^*(x, y)$ is active which makes the governing energy equations linear. The temperature field due to each active heat layer may be computed separately and then added linearly to provide the temperature field due to all n heat sources [9]. $T_{i,i^*}(x, y, z_i)$ is the steady-state temperature rise in the i th die due to heat generation $Q_i(x, y)$ in the i *th die. For brevity of nomenclature, the subscript i^* is omitted henceforth, and the temperature field in this case is represented as $T_{i,i}(x, y, z_i)$. Governing energy equations along with the boundary conditions are programmed and solved separately for each die.

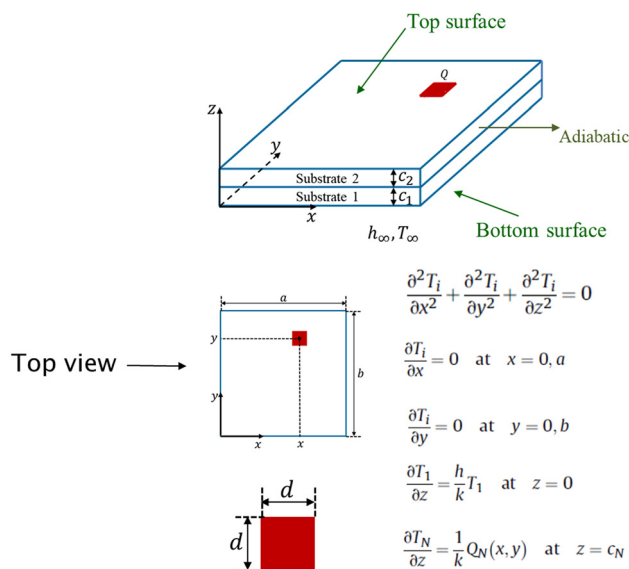


Fig. 1 Heat transfer description of the problem

Table 1 Dimensions of the heat source and the substrate shown in Fig. 1

SI no	Dimension	Value (mm)
1	a	10
2	b	10
3	C_1	0.5
4	C_2	0.5
5	D	1

$q_i(x, y)$ is the heat flux [9] entering the i th die at $z_i = c_i$. k_i is the thermal conductivity of the i th die. $T_{0i}(x, y)$ is the temperature profile at the intersecting plane between die i and $i - 1$.

Assuming constant thermal conductivity, each $T_i(x, y, z_i)$ satisfies the following governing steady-state energy equation and adiabatic boundary conditions [9]:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0 \quad (1)$$

Boundary conditions in x and y directions are as follows:

$$\frac{\partial T}{\partial x} = 0, \quad \text{at } x = 0, a \quad (2)$$

$$\frac{\partial T}{\partial y} = 0, \quad \text{at } y = 0, b \quad (3)$$

Boundary conditions in z -direction is as follows:

$$k \frac{\partial T_2}{\partial z} = q_2(x, y) \quad \text{at } z = c_2 \quad (4)$$

$$k \frac{\partial T_1}{\partial z}(\text{at } z = c_1) = k \frac{\partial T_2}{\partial z}(\text{at } z = 0) + q_1(x, y) \quad (5)$$

$$\frac{\partial T_1}{\partial z}(\text{at } z = 0) = \frac{h}{k} T_1(\text{at } z = 0) \quad (6)$$

A detailed procedure on the analytical solution is well explained in Ref. [9].

The analytical code was validated by a detailed computational fluid dynamics model using commercially available tool ANSYS FLUENT [8] and the maximum error between the code and computational fluid dynamics was found to be less than 8% (for uniformity) and less than 2% for maximum temperature.

Unique Geometric Parameter λ Definition

A unique parameter λ is defined which takes into account the closeness of the heat sources to center on any layer and the closeness of the heat sources to each other on any layer.

The definition of λ 's is as follows:

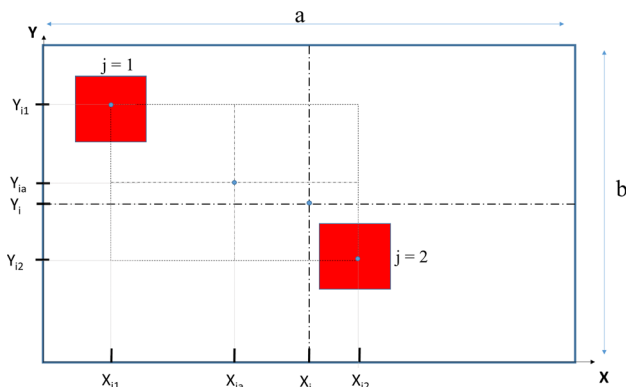
λ_1 and λ_3 defines the distance between the centroid of the heat source to the centroid of the arrangement in layers 1 and 2, respectively,

$$\lambda_1 = (X_{ij} - X_{ia})^2 + (Y_{ij} - Y_{ia})^2 \quad (7)$$

λ_2 and λ_4 defines the distance between the centroid of the substrate to the centroid of the arrangement in layers 1 and 2, respectively,

$$\lambda_2 = (X_i - X_{ia})^2 + (Y_i - Y_{ia})^2 \quad (8)$$

X_{ij} , Y_{ij} denotes the centroid of each heat sources and X_{ia} , Y_{ia} denotes the mean of the centroids of two heat sources (arrangement) in a layer; X_i and Y_i denotes the centroid of the substrate of

**Fig. 2 Schematic representation of calculation of geometric parameters in any layer for two heat sources ($j = 1$ and 2) (Eqs. (7) and (8))**

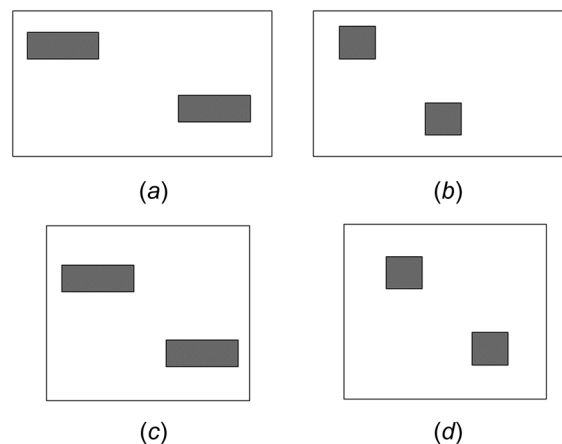
i th layer; X_{1c} and Y_{1c} denote the centroid of the substrate layer 1 as shown in Fig. 2.

The physical significance of such a definition of λ 's (Lambdas) would help in optimization of the heat sources for any aspect ratio of the heat sources and substrate. For the problem under consideration square heat sources and square substrates are chosen for demonstration. Figure 3 depicts the possible combinations of heat sources and the substrate shapes for which the geometric parameter λ can represent the position of the heat sources. This paper demonstrates the optimization for the configuration shown in Fig. 3(d).

The depiction of the calculation of centroid of each heat source, substrate and arrangement is shown in Fig. 4.

Furthermore, to understand the significance of λ , the limiting (extreme) values are examined:

- $\lambda_1, \lambda_3 \rightarrow 0$: The heat sources are placed closer to each other (shown in Fig. 5).
- $\lambda_1, \lambda_3 \rightarrow \infty$: The heat sources are crowded toward the corner of the substrate and away from each other.
- $\lambda_2, \lambda_4 \rightarrow 0$: The heat sources are having symmetrical arrangement.
- $\lambda_2, \lambda_4 \rightarrow \infty$: The heat sources are moving toward the corner of the substrates and closer to each other (shown in Fig. 6). Furthermore, there is a relationship between the maximum possible λ_1 for each value of λ_2 as depicted in Fig. 7.

**Fig. 3 Schematic of the possible shapes of the substrates and chip: (a) rectangular chips on rectangular substrates, (b) square chips on rectangular substrates, (c) rectangular chips on square substrates, and (d) square chips on square substrates**

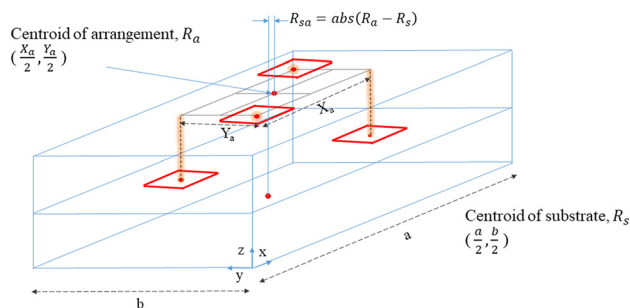


Fig. 4 Schematic of the centroids of the arrangement of heaters

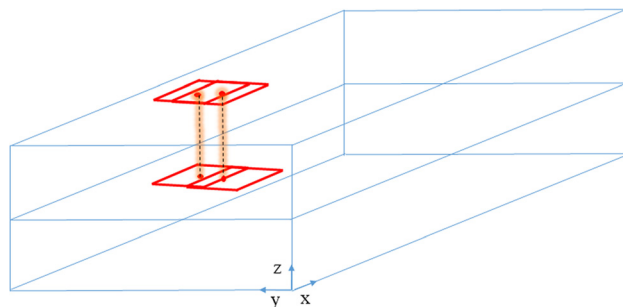


Fig. 5 Schematic of the heater arrangement for very low values of λ_1 and λ_3

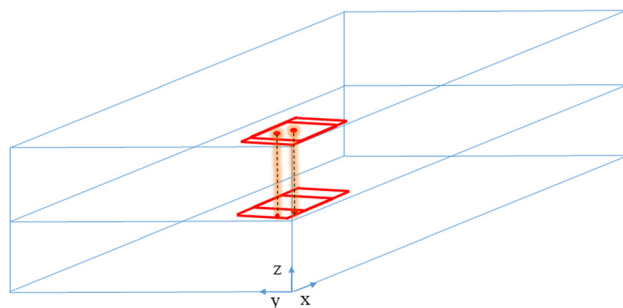


Fig. 6 Schematic of the heater arrangement for very high values of λ_2 and λ_4

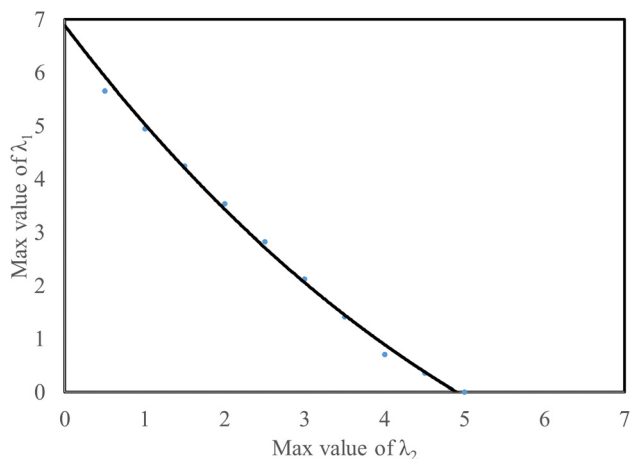


Fig. 7 Plot showing the relationship between λ_1 and λ_2

Latin Hypercube Sampling

In order to generate a reasonable number of samples of different positions of the heat sources uniquely across the two layers, Latin hypercube sampling (LHS) was employed with the spatial constraints. The sampling algorithm was self-adaptive and self-learning with three stages. The variables were nondimensionalised to eliminate the effects of substrate and chip geometry. The usage of LHS over other random sampling or orthogonal sampling techniques is justified by careful assessment [10] and other techniques used for selecting initial designs. Sampling a function of 16 variables, the range of each variable is divided into T equally probable intervals. T sample points are later placed to strictly meet the Latin hypercube requirements. This constraint forces the number of divisions, T , to be equal for each input variable. The LHS scheme does not require more samples for more dimensions (variables). Srikanth and Balaji [17], used LHS to generate a plausible collection of parameters to train the artificial neural network for optimization of phase-change material-based matrix type composite heat sink.

In this study, three stages of the self-learning sampling algorithm is depicted in Fig. 8. The learning in each stage was achieved by Brute-Force (exhaustive search method).

The stage 1 sampling algorithm learned that the λ_1 and $\lambda_3 = 0$ are highly undesirable and such cases were eliminated for the stage 2 sampling. The stage 2 sampling algorithm learned that the λ_2 and $\lambda_4 = \infty$ are highly undesirable and such cases were eliminated for the stage 3 sampling.

For the given adiabatic boundary condition on the four walls and constant heat transfer coefficient boundary condition on the top wall, λ_2 and $\lambda_4 = 0$ turns out to be the most optimal configuration. It can be inferred from close observation of definition of λ that a symmetrical arrangement of heat sources about substrate centroid is only possible when λ_2 and λ_4 becomes equal to zero. A small deviation from this value would lead to nonsymmetrical arrangement of heat sources. But it is highly impossible to determine the value of λ_1 for such configuration and to find the exact location of the chip. Henceforth for the stage 3 optimization, definition of R and θ is defined and a functional relationship with the

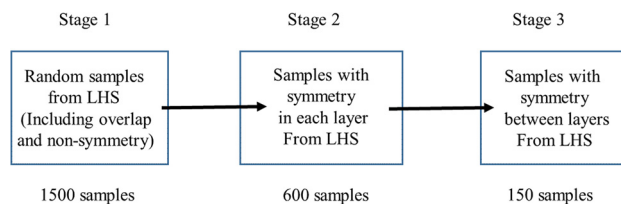


Fig. 8 Stages of the self-learning sampling algorithm

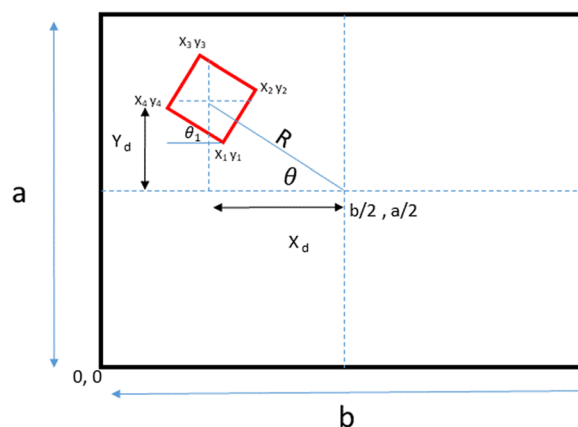


Fig. 9 Schematic representation of R and θ

response parameters ($T_{\max}$ and σ) is desired to be established. The depiction of R and θ is shown in Fig. 9 ($\theta_1 = 0$ in this paper). Each value of R and θ can define an arrangement when λ_2 and $\lambda_4 = 0$.

The R and θ were calculated using the equations

$$R = \sqrt{X_d^2 + Y_d^2} \quad (9)$$

$$\theta = \tan^{-1} \left(\frac{Y_d}{X_d} \right) \quad (10)$$

$$R_{\max} = \sqrt{\left(\frac{b}{2} - \frac{\text{chip length}}{2} \right)^2 + \left(\frac{a}{2} - \frac{\text{chip width}}{2} \right)^2} \quad (11)$$

$$R_{\min} = 0 \quad (12)$$

$$R^* = \frac{R - R_{\min}}{R_{\max} - R_{\min}} \quad (13)$$

$$\theta_{\max} = 90 \text{ deg} \quad (14)$$

$$\theta_{\min} = 0 \text{ deg} \quad (15)$$

$$\theta^* = \frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}} \quad (16)$$

Regression

The regression was carried out using the state of the art Levenberg–Marquadt algorithm for the nonlinear least square regression. A higher-order polynomial functional form was chosen. The final functional form for the maximum temperature and temperature uniformity on the convective surface as a function of chip location is as follows:

$$T_{\max} = 250 \times (0.83 - 1.43R^* + 2.06R^{*2} - 0.73R^{*3} - 0.6\theta^* + 0.6\theta^{*2}) \quad (17)$$

$$\sigma = 65.13 - \frac{65}{R^*} + \frac{28.6}{R^{*2}} - \frac{5.3}{R^{*3}} + \frac{0.4}{R^{*4}} - \frac{9.4}{R^{*5}} - 35.8\theta^* + 35.6\theta^{*2} + 0.31\theta^{*3} \quad (18)$$

The value of coefficient of regression is 0.94 and the parity plot for the regression is shown in Fig. 10.

The objective functions for optimization are also derived from ANN, specifically in this case from the output of a feed-forward back propagation ANN. ANN is a nonlinear regression tool inspired by the structure of biological neural networks. ANNs consist of three types of layers—input, hidden, and output [10]. The input layer consists of nodes, with each node corresponding to an input variable. The number of neurons in the hidden layer is variable and results in different models. The results of the neuron-independent study is shown in Fig. 11. The task of the artificial neural network is to calculate the response parameters for a given set of input variables. The artificial neural network must accurately calculate the output as we are using a supervised learning technique. Generally, the sigmoid activation function and the number of hidden neurons/layers are assumed, and the weights for the network are calculated in such a way that the error of prediction is minimized.

Some of the training algorithms used are the feed-forward back propagation algorithm and the Levenberg–Marquadt algorithm. There are three steps involved in these algorithms, namely, training, validation, and testing. Training calculates the weights of the

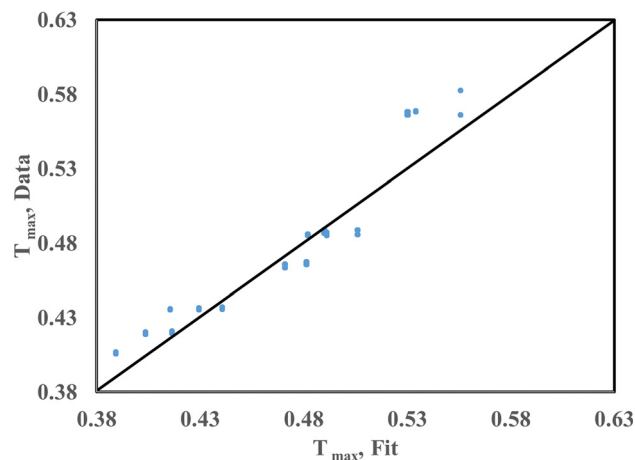


Fig. 10 Parity plot for the regression

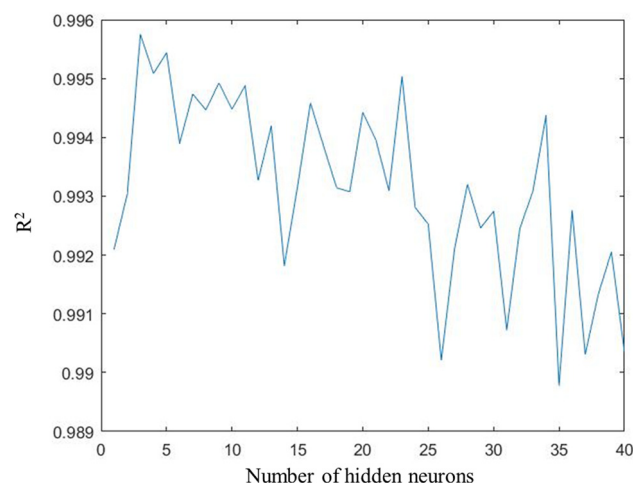


Fig. 11 Variation of regression coefficient with the number of hidden neurons

network in such a way that the error of the training data is in minimized. The validation is employed for dataset that was not a part of the network training to ensure robustness of the network's interpolation. Testing calculates the error in the final network generated and confirms the predictive power of the network. The back propagation algorithm is implemented, to adjust the weights to minimize the error between the neural network predicted output and desired output. Parity plot for each stage of ANN is as shown in Fig. 12. The coefficient of regression value is so close to one which confirm the high quality of the regression (note that the coefficient of regression is denoted as " R^2 " and this has no relation with R that is defined as the heater location).

Optimization

Any optimization problem is driven by constraints more than the objective function. A better understanding of the constraints lead to better optimal solutions. The constraints are derived from the results of stages 1 and 2 of the sampling algorithm. The two main objective of the optimization are to minimize the temperature gradient, (σ) and minimize the maximum temperature of the heat source. The optimization problem can be mathematically stated as

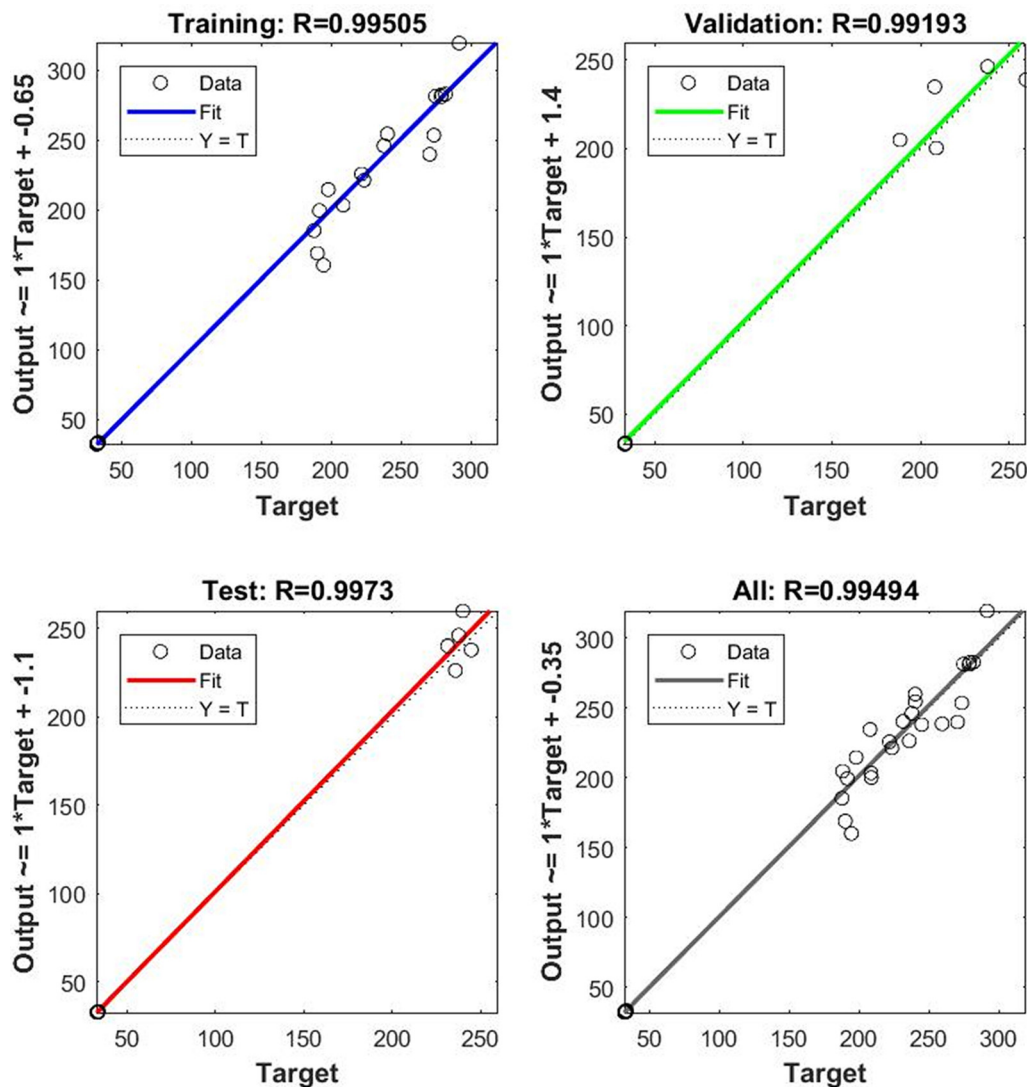


Fig. 12 Parity plot from neural network

$$F1: \text{minimize } \sigma = f(R^*, \theta^*, \lambda_1, \lambda_2, \lambda_3, \lambda_4) \quad (19)$$

$$F2: \text{minimize } T_{\max} = f(R^*, \theta^*, \lambda_1, \lambda_2, \lambda_3, \lambda_4) \quad (20)$$

Subject to

$$\lambda_2 = 0 \quad (21)$$

$$\lambda_4 = 0 \quad (22)$$

$$0 \leq R^* \leq 1 \quad (23)$$

$$0 \leq \Theta^* \leq 1 \quad (24)$$

One more constraint is that the heat sources are at least 1 mm apart from each other on both the layers. From the heat transfer physics, it is also quite intuitive that the interaction effect between the two layers also plays an important role in the response parameters. No two heaters from each layer should be exactly on top of each other. Violating this constraint produces high temperature and low uniformity. This was confirmed with the numerical simulations for the cases shown in Fig. 13 and the corresponding results are shown in Table 2.

In the early stage of optimization process the conflicting nature of the objectives were evaluated. Additionally, the initial cases

shown in Fig. 13 were validated again by ANSYS FLUENT and the maximum discrepancy was found to be less than 8.3% (overlapping cases) for uniformity and 2% for maximum temperature. The average discrepancy between the analytical solutions and ANSYS FLUENT was less than 3.2%. From Fig. 14, it is evident that the objectives are not conflicting much (throughout the variable

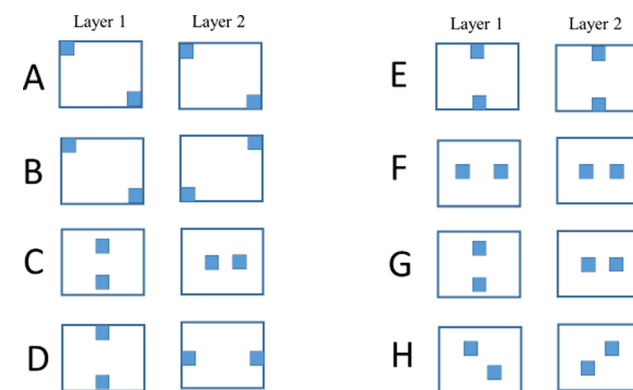


Fig. 13 Preliminary cases evaluated to decide on the constraints

Table 2 Response parameters for the cases considered in Fig. 2

Case ID	σ	$T_{\max}$ (°C)
A	26	258
B	15	171
C	7.4	126
D	9.4	133
E	15	171
F	12.5	142
G	8.5	132

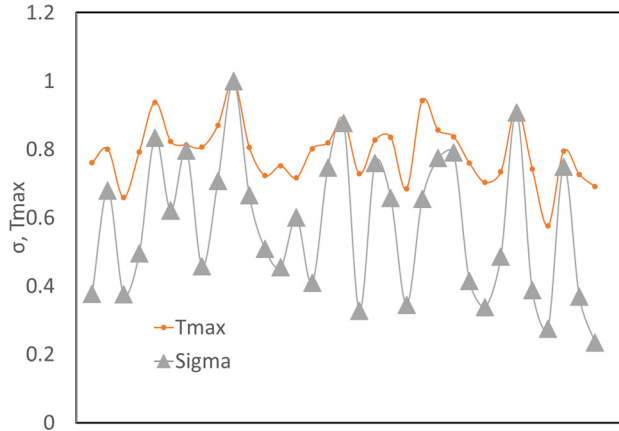


Fig. 14 Nonconflicting nature of the objectives obtained from the analytical solution

search space generated from LHS). Therefore, a multi-objective optimization would not be thoroughly applicable, rather optimizing any one of the objectives, optimizes the other. The ANN model and a polynomial regression model is coupled with genetic algorithm to obtain the optimal value. The flowchart for optimization is as depicted in Fig. 15.

Genetic Algorithm

Genetic algorithms (GAs) are numerical optimization algorithms inspired from evolutionary phenomena like the natural selection. Genetic algorithms are commonly used to determine the global optimum in the feasible domain. The ANN trained is used to serve as the fitness function for the genetic algorithm-based optimization procedure to obtain the optimum location of the heat sources. GA is one among a large class of evolutionary-based algorithms. Each candidate solution has unique properties, which can be altered by mating to yield better candidate solution and are represented in binary as strings of 0s and 1s.

The evolution is an iterative process starting with selection of a random population and the population in each iteration is called a generation. For every generation, the fitness of every individual in the population is evaluated. The fitness is the value of the objective function for the optimization problem. GAs always try to improve the fitness with each generation. From the current generation, the fittest individuals are selected and their genetics are modified to generate next generation. These individuals move to the next generation of the algorithm. This process continues until either the maximum number of generations is reached or a satisfactory fitness level is attained. The strong ability of the genetic algorithms to tackle the search spaces containing many local optima is one of its attractive features for engineers. In contrary to the conventional search methods and calculus, genetic algorithms initiates the search at multiple number of points in the search space. The desired accuracy in the value of “ R^* ” and “ θ^* ” is

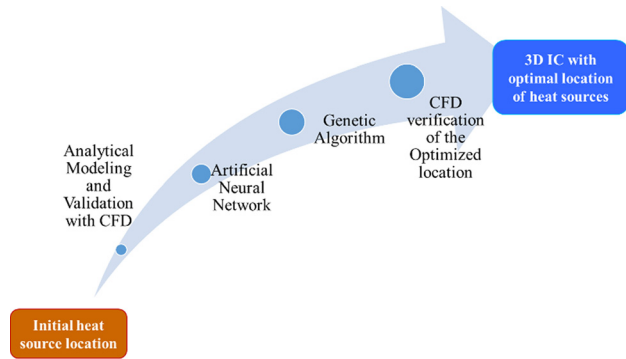


Fig. 15 Optimization approach

chosen to be two. This decides the binary string length of the variables. Initial population is generated randomly with a size of 30. Once the initial population is generated the corresponding fitness value is evaluated and ranking is done. Subsequently operators like selection, crossover, and mutation are applied to the population till the optima is obtained. In this study, the new individuals are generated using two parent two point crossover. Roulette wheel random selection method is used to identify the two possible mating strings to produce the next generation. These mating strings would undergo crossover only if a random number is greater than or equal to the probability of crossover. A two point crossover is utilized, the two positions of the binary strings at which the crossover exchange between these mating strings occur, also depends on the random number generated. Mutation is applied to all the individuals provided the random number generated is greater than or equal to the probability of mutation. The newly generated binary strings are converted to integers and then to corresponding real number. The population number (N), generation/iteration count (G), probability of crossover (P_c), and probability of mutation (P_m) have been arrived at after an extensive study. Finally, the parameters $N=40$, $G=200$, $P_c=0.6$, $P_m=0.4$ is arrived after an extensive sensitivity study.

Results and Discussion

The discussion pertains to minimizing the temperature gradient which is defined as the coefficient of variation of the temperature on the top convective surface (σ).

Standard deviation can be obtained as

$$s = \sqrt{\frac{\sum_{i=1}^N (T_i - \bar{T})^2}{N-1}} \quad (25)$$

N denotes the number of grid points at which the temperature measurement was recorded. $N=121$ in this study. T_i represents the value of temperature at each point of the grid

$$\bar{T} = \frac{\sum_{i=1}^N T_i}{N} \quad (26)$$

$$\sigma = \frac{s}{\bar{T}} \quad (27)$$

Minimize σ . The genetic algorithm was coupled with the ANN to derive the objective function to minimize σ as a function of the heat source location. The optimal configuration that was obtained is as shown in Fig. 16. It is evident from the Fig. 16 that the heat sources are arranged symmetrically in both the layers and they are at an optimal (distance beyond or below which the temperature

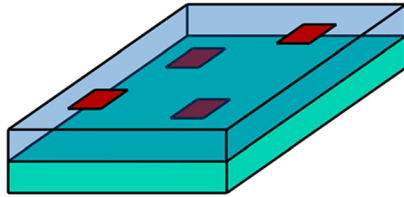


Fig. 16 Isometric view of the optimal arrangement

Table 3 Optimal values of the parameters

Parameter	Optimal value
R	3.53
θ	0.785
λ_1	5
λ_2	0
λ_3	5
λ_4	0

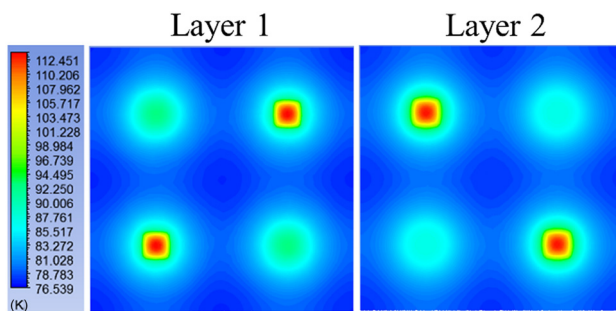


Fig. 17 Temperature contours in layers 1 and 2 of the optimal arrangement

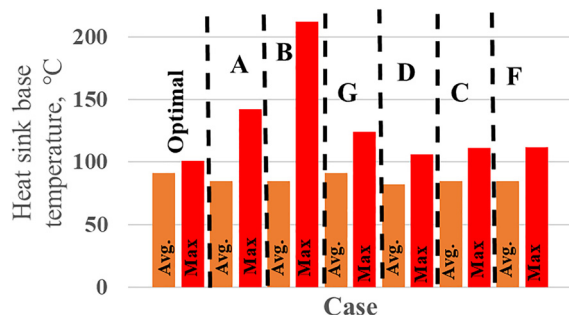


Fig. 18 Comparison of heat sink base temperature for cases explained in Fig. 2

nonuniformity increases) distance from the walls and from the center.

The optimal value of the parameters for the problem under consideration is listed in Table 3. From heat transfer physics, it is also quite intuitive that a symmetrical arrangement of heat sources are desired for better spreading of the heat from both the layers. When the λ_1, λ_3 is increased, the heat sources move closer to the adiabatic walls and the maximum temperature on the substrate increases. Furthermore, if the λ_1, λ_3 are reduced, the heat sources move closer to each other and the heat sources interaction effect increases, allowing the chip temperature to rise. The optimal value of R and θ defines the optimal location of the chip. The value of σ

for this configuration was found to be 4.12 (evident from Fig. 18) and the maximum die temperature corresponds to a $T_{\max}$ of 110.

Figure 17 shows the temperature contours on the top layer for the optimal configuration. It is evident that the spreading is more uniform which results in minimum hot spot temperature too. Isothermal conditions are preferred to have an accurate estimate of the heat sink thermal resistance based on a single temperature. The optimal case thus derived in this study has a near isothermal surface as seen from Fig. 18 the discrepancy between the average and the maximum temperature is minimum. Calculation of thermal resistance using the average or the maximum temperature do not show significant difference.

Furthermore, the optimal arrangement of the heat sources could also ensure uniform temperature at the convective surface and thereby maintaining the chip temperature within limits for a 3D ICs.

Conclusions

A robust multi-objective constrained optimization was performed for a two layer two hotspot case by using artificial neural network coupled with genetic algorithm in a vertically stacked two layer 3D-ICs. The shape of the substrate and the chip were considered to be squares. A polynomial function and a feed-forward back propagation neural network was developed that represents the function relationship between the position of the heaters and the temperature uniformity. Two objectives, namely, minimizing the temperature gradient (σ) and the maximum temperature at convective surface was considered with a set of constraints. The nonconflicting nature of the objectives reduced the multi-objective problem to single objective. The results of the optimization revealed that the most uniform temperature distribution is achieved by having the heat source locations corresponding to $\lambda_2, \lambda_4 = 0$ on both the layers.

The major conclusions from this study are as follows:

- Optimizing for the maximum surface temperature eventually optimized the temperature uniformity.
- A self-learning sampling algorithm was designed to understand the physics of the problem in each stage of learning.
- There exist an optimal location ($R = 3.53$ and $\theta = 0.785$) of the heaters that minimized the discrepancy between the maximum and average temperature on the convective surface, thereby increasing uniformity.

Nomenclature

- G = number of generations
 N = size of the initial population
 P_c = probability of crossover
 P_m = probability of mutation
 R^2 = regression coefficient
 R^*, θ^* = nondimensional position of the heater
 T = temperature, °C
 X = centroid of heat source/substrate/arrangement in X direction, mm
 X, Y = centroid
 Y = centroid of heat source/substrate/arrangement in Y direction, mm
 λ = geometric parameter of heat source location, mm²
 σ = nondimensional temperature gradient

Subscripts

- a = arrangement
 i = layer/substrate
 j = heat source

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